

Frederik Maximilian Willemsen



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Education

Technical University Munich

Master of Science in Computer Science

University of California, Davis

Exchange semester – GPA: 3.92/4.0

Technical University Munich

Bachelor of Science in Computer Science – Grade Distinction: Passed with merit

Oct. 2025 – Dec. 2027

Munich, Germany

Sep. 2023 – Jan. 2024

Davis, California

Oct. 2021 – Aug. 2025

Munich, Germany

Experience

TUM.ai

Software Engineer

May 2026 – now

Munich, Germany

- Europe's leading AI student initiative. Building a member management platform to centralize operations across 400+ members and automating finance workflows to eliminate recurring manual overhead for the organization.

Deloitte Consulting

AI & Data Project Intern | Python, Flask, Angular, Pandas, SQL, AWS, Excel, Powerpoint

Oct. 2025 – Feb. 2026

Munich, Germany

- Developed and deployed to production an agentic AI chatbot on AWS, built on a multi-agent architecture leveraging a RAG pipeline with source citations and full-stack implementation (frontend, backend, APIs) Presented by Senior Management at an international Deloitte AI conference in Japan to acquire new enterprise clients.
- Built and trained a machine learning prediction model for customer lead prioritization in automotive aftersales, measurably improving efficiency in aftersales lead management.
- Selected for Deloitte's Top 10% Talent Pool based on outstanding internship performance.

Check24

Software Engineer Intern | Java, SpringBoot, React

Aug. 2025 – Oct. 2025

Munich, Germany

- Designed and deployed an MCP server for structured LLM context delivery, reducing API latency by 50% via caching, improving context reliability and lowering costs.
- Designed, implemented, and shipped to production an AI-powered natural-language filtering system serving 15 million+ users, enabling intuitive free-text search and automatic filter application on Check24's live platform.

Projects

Data-Driven Selection of Algorithmic Configurations

Bachelor's Thesis | C++, Python, Pandas

Mar. 2025 – Aug. 2025

TUM: Chair of Physics-Enhanced Machine Learning

- Designed and implemented a machine-learning-based tuning algorithm to optimize algorithmic configurations in high-performance particle simulations at the chair of physics-enhanced machine learning at TUM.
- Achieved a 16% runtime speedup compared to legacy approaches on a large-scale HPC system (CoolMuc-4).
- Applied Random Forests, K-Means, PCA, and performance-driven experimentation to enable data-driven decision-making in simulation workflows.

Scientific Computing (PSE) Molecular Dynamics

Practical Course Project | C++, OpenMP

Oct. 2024 – Feb. 2025

Technical University Munich

- Led a 3-person team in a practical course with 8 competing teams to develop a high-performance molecular dynamics simulation framework. Achieved the best overall performance, delivering a 60% speedup over the next-best team through optimized parallelization and low-level optimizations.

TUM.ai Agora Hacks Hackathon

agorix.eu | TypeScript, Python

10 – 12 Apr. 2026

Munich, Germany

- Developed agorix.eu in under 48 hours, an AI-powered civic technology platform that helps Munich residents navigate fragmented government information, leverages LLMs to aggregate, synthesize, and contextualize real-time data from official municipal systems into actionable, citizen-friendly insights.

Technical Skills & Extracurriculars

Languages: Java, Python, C++, C, SQL

Frameworks & Tools: Spring Boot, Flask, React.js, Angular, AWS, PyTorch, Pandas, OpenMP

Leadership: Student Council President, Board Member student housing organization, Member Uganda Club

Hobbies: Half-marathon runner, Golf index: 2.0, Football